B.TECH 1ST YEAR | ENGINEERING CHEMISTRY

Surface Chemistry & Nanomaterials

Unit V – Complete Study Module

Adsorption · Colloids · Nanomaterials · Isotherms · Applications

Adsorption Isotherms (Freundlich, Langmuir, BET)

Colloidal Systems & Micelle Formation

Nanoparticle Synthesis & Stabilization

Industrial & Medical Applications

Exam-Oriented · Diagram-Rich · Q&A Included

Prepared as per VTU / Anna University B.Tech Curriculum |

Table of Contents

How to use this module: Read theory → Study diagrams → Memorise formulas → Practise Q&A → Revise using Quick Revision Sheet before exam.

1 Introduction to Surface Chemistry

1.1 Definition

Surface Chemistry is the branch of chemistry that deals with phenomena occurring at the interfaces (surfaces) between different phases — solid–liquid, solid–gas, liquid–gas, and liquid–liquid. It studies how molecules behave at boundaries and how surface forces influence physical and chemical processes.

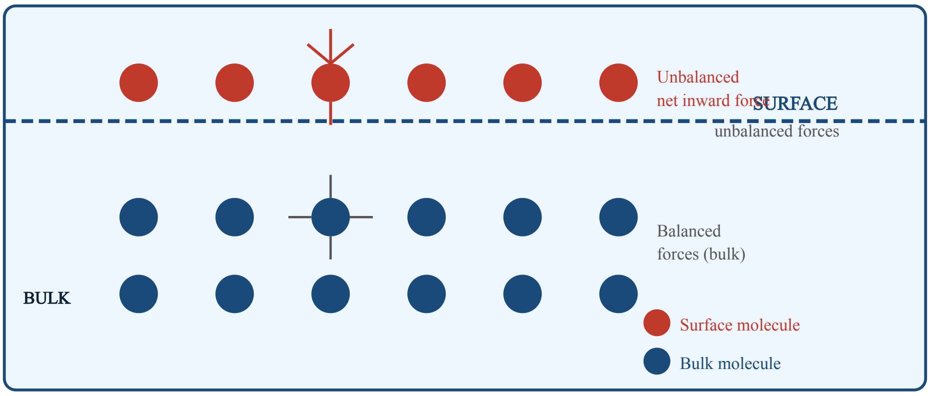
1.2 Importance and Scope

Surface chemistry governs many natural and industrial processes. Catalysis, corrosion, detergency, drug delivery, and semiconductor fabrication all rely fundamentally on surface interactions. The surface area available for interaction is inversely related to particle size — hence nanoparticles have enormous surface-to-volume ratios.

1.3 Surface Energy

Molecules at a surface experience unbalanced forces compared to bulk molecules. This excess energy is called **surface energy** (or surface tension for liquids). Systems tend to minimise surface energy, leading to phenomena like droplet formation, adsorption, and wetting.

FIG 1.1 – SURFACE VS. BULK MOLECULAR ENVIRONMENT



Surface molecules have unbalanced intermolecular forces leading to surface energy

1.4 Key Terms

Term	Meaning
Interface	Boundary between two phases (e.g., solid-liquid)
Surface tension	Inward force per unit length at a liquid surface
Wetting	Spreading of a liquid over a solid surface
Adsorption	Accumulation of substances at a surface

Absorption	Uniform penetration into bulk of a substance

Remember (Exam Tip): Adsorption is a *surface* phenomenon; absorption is a *bulk* phenomenon. A sponge absorbs water; silica gel adsorbs moisture.

2 Adsorption and Its Types

2.1 Definition of Adsorption

Adsorption is the process by which molecules or ions from a gas, liquid, or dissolved solid accumulate on the surface of a solid (or less commonly liquid), forming a thin film. The solid that adsorbs is the **adsorbent**; the substance adsorbed is the **adsorbate**.

2.2 Types of Adsorption

Adsorption is classified into two main types based on the nature of forces involved:

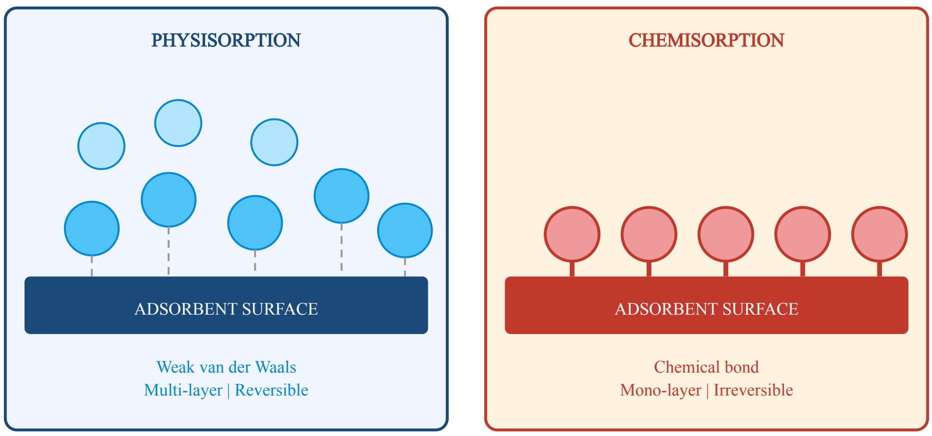
A. Physical Adsorption (Physisorption)

- Caused by weak **van der Waals forces** between adsorbent and adsorbate.
- Reversible; adsorbate can be easily desorbed by decreasing pressure or increasing temperature.
- Low enthalpy of adsorption: 5–40 kJ/mol.
- Multilayer adsorption possible.
- Occurs at low temperatures; favoured by high pressure.
- **Example:** N_2 adsorption on silica gel.

B. Chemical Adsorption (Chemisorption)

- Involves formation of **chemical bonds** (covalent or ionic) between adsorbate and adsorbent.
- Irreversible or difficult to reverse.
- High enthalpy: 40–400 kJ/mol.
- Monolayer adsorption only.
- Occurs at higher temperatures; specific to particular gas-solid pairs.
- **Example:** H_2 adsorption on Ni catalyst surface.

FIG 2.1 – PHYSISORPTION VS. CHEMISORPTION



Comparison of Physisorption (multilayer, weak) and Chemisorption (monolayer, strong bonds)

2.3 Comparison Table

Property	Physisorption	Chemisorption
Force	van der Waals	Chemical bond
Reversibility	Reversible	Irreversible
Enthalpy (kJ/mol)	5 – 40	40 – 400
Temperature	Low temp. favoured	High temp. favoured
Layers	Multi-layer	Mono-layer
Specificity	Non-specific	Highly specific

2.4 Factors Affecting Adsorption

- **Nature of adsorbent:** Porous materials (activated charcoal, silica) have greater surface area.
- **Surface area:** Greater surface area → more adsorption.
- **Temperature:** Physical adsorption decreases with temperature; chemisorption shows a maximum.
- **Pressure:** Adsorption increases with pressure up to a saturation point.
- **Nature of adsorbate:** Easily liquefiable gases (high critical temperature) are adsorbed more.

3 Colloids – Basics and Properties

3.1 Definition

A **colloid** (colloidal dispersion) is a two-phase heterogeneous system in which one substance (the **dispersed phase**) is dispersed in another substance (the **dispersion medium**) in the form of very small particles ranging from **1 nm to 1000 nm (1 µm)** in size.

3.2 Classification of Dispersions

Property	True Solution	Colloid	Suspension
Particle size	< 1 nm	1–1000 nm	> 1000 nm
Visibility	Invisible	Ultramicroscope	Visible
Tyndall effect	Absent	Present	Absent
Filterability	Passes filter	Passes filter	Retained
Stability	Very stable	Moderately stable	Unstable
Example	NaCl in water	Milk, blood	Chalk in water

3.3 Types of Colloids

Based on the state of dispersed phase and dispersion medium:

Dispersed Phase	Medium	Type	Example
Solid	Liquid	Sol	Paint, blood, starch sol
Liquid	Liquid	Emulsion	Milk, mayonnaise
Gas	Liquid	Foam	Whipped cream
Solid	Gas	Aerosol	Smoke, dust
Liquid	Gas	Aerosol	Fog, mist
Solid	Solid	Solid sol	Alloys, gem stones

3.4 Properties of Colloids

A. Tyndall Effect

When a beam of light passes through a colloidal solution, the path of light becomes visible due to **scattering of light** by colloidal particles. This is the Tyndall effect. It is used to distinguish colloids from true solutions.

B. Brownian Motion

Colloidal particles show random, zig-zag movement due to unequal bombardment by medium molecules. This is called **Brownian motion**, first observed by Robert Brown. It prevents sedimentation and helps maintain stability.

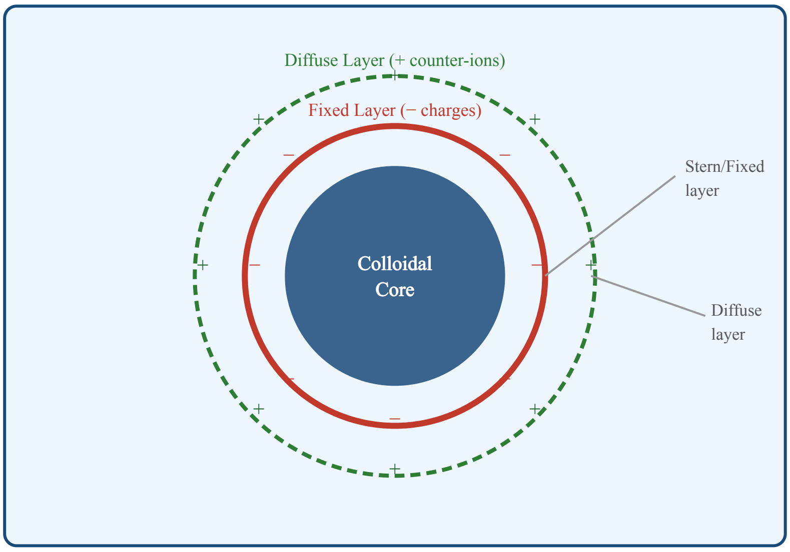
C. Electrophoresis

Migration of colloidal particles under an applied electric field. Particles move toward the electrode of opposite charge. Used to determine charge on particles.

D. Coagulation / Flocculation

Addition of electrolytes neutralises the charge on colloidal particles, causing them to aggregate and settle. This is coagulation. **Hardy–Schulze rule:** Higher the valency of the oppositely charged ion, greater its coagulating power.

FIG 3.1 – STRUCTURE OF A COLLOIDAL PARTICLE (DOUBLE LAYER)



Electric double layer around a negatively charged colloidal particle

4 Micelle Formation

4.1 Definition

A **micelle** is an aggregate of surfactant (surface-active agent) molecules, spontaneously formed in a solution above a critical concentration called the **Critical Micelle Concentration (CMC)**. In water, micelles have a hydrophobic core and hydrophilic exterior.

4.2 Structure of Surfactant Molecule

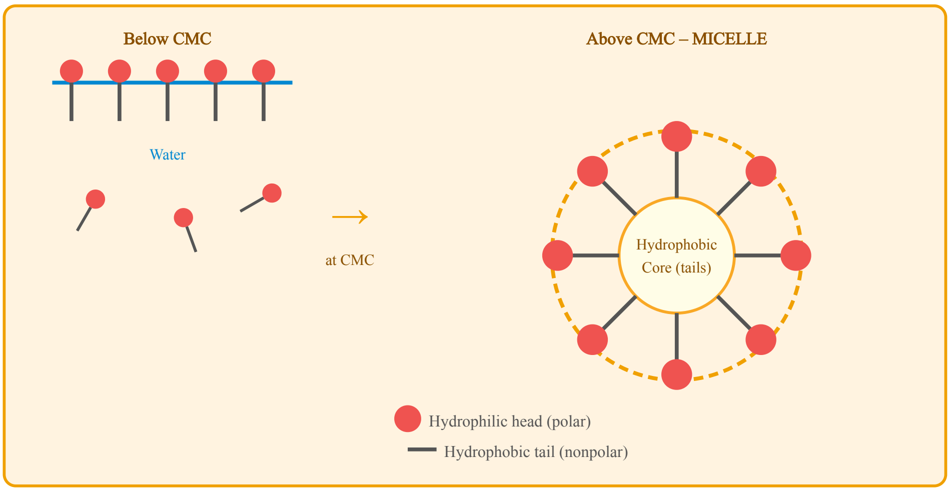
A surfactant molecule has two distinct parts:

- **Hydrophilic head** – polar group that is attracted to water (e.g., $–COO^-$, $–SO_3^-$, $–NH_4^+$).
- **Hydrophobic tail** – long nonpolar hydrocarbon chain that repels water.

4.3 Mechanism of Micelle Formation

1. **Below CMC:** Surfactant molecules spread at the water surface as a monolayer with tails pointing away from water.
2. **At CMC:** The surface becomes saturated. Additional molecules aggregate into micelles.
3. **Above CMC:** Spherical micelles form with tails pointing inward (hydrophobic core) and heads facing water (hydrophilic shell).
4. The CMC for soap (sodium stearate) is about 10^{-3} mol/L.

FIG 4.1 – MICELLE FORMATION DIAGRAM



Surfactant arrangement at surface (below CMC) and spherical micelle formation (above CMC)

4.4 Applications of Micelles

- **Detergency:** Grease/oil trapped in hydrophobic core, washed away with water.
- **Drug delivery:** Hydrophobic drugs encapsulated in micelle core for targeted delivery.
- **Emulsification:** Stabilisation of oil-water emulsions.

- **Nanoparticle synthesis:** Reverse micelles used as nanoreactors.

5 Synthesis of Colloids (Bragg's Method)

5.1 Methods of Preparation of Colloids

Colloids can be prepared by two approaches:

- **Dispersion methods** – breaking down bulk material into colloidal size (top-down).
- **Condensation methods** – building colloidal particles from smaller units (bottom-up).

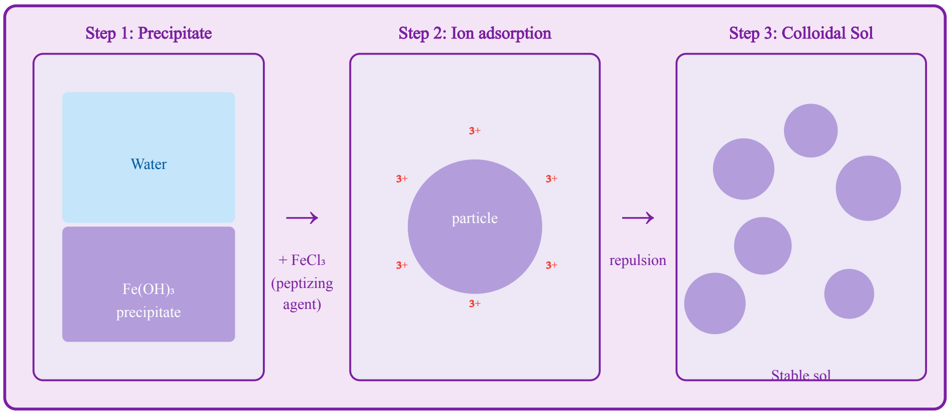
5.2 Bragg's Method (Peptization)

Peptization is the process of converting a freshly prepared precipitate into a colloidal solution by adding a small amount of electrolyte called the **peptizing agent**. The ions of the electrolyte are adsorbed on the precipitate particles, giving them a charge. Electrostatic repulsion breaks the precipitate into colloidal particles.

Steps in Peptization

1. A freshly prepared precipitate is taken (e.g., $Fe(OH)_3$).
2. A small quantity of peptizing agent is added (e.g., $FeCl_3$ solution).
3. Fe^{3+} ions are adsorbed on $Fe(OH)_3$ particles, giving them positive charge.
4. Electrostatic repulsion causes particles to break apart into colloidal size.
5. A stable reddish-brown $Fe(OH)_3$ sol is formed.

FIG 5.1 – PEPTIZATION (BRAGG'S METHOD) – COLLOIDAL SOL FORMATION



Peptization: $FeCl_3$ peptizes $Fe(OH)_3$ precipitate into a stable colloidal sol

5.3 Other Preparation Methods

Method	Principle	Example
Bredig's Arc Method	Electric arc between metal electrodes in water	Au, Ag, Pt sols
Chemical Reduction	Reduction of metal salts	Gold sol (Faraday method)
Double Decomposition	Metathesis reaction	As_2S_3 sol

Ultrasonic Dispersion	High-frequency sound waves	Mercury sol
Peptization	Adsorption of ions on precipitate	$Fe(OH)_3$ sol

5.4 Purification of Colloids

- **Dialysis:** Removal of crystalloid impurities using a semipermeable membrane.
- **Electrodialysis:** Dialysis under applied electric field.
- **Ultrafiltration:** Filtration through ultra-fine membranes.

6 Stabilization of Colloids

6.1 Why Stabilization is Needed

Colloidal particles tend to aggregate (coagulate) due to van der Waals attractive forces. Stabilization prevents aggregation and maintains the colloidal state. The principal mechanisms are:

- **Electrostatic repulsion** – surface charge on particles repels like-charged particles.
- **Steric stabilization** – polymer chains adsorbed on particle surface create a physical barrier.

6.2 Role of Charge (Zeta Potential)

The **zeta potential (ζ)** is the electric potential at the boundary between the fixed and diffuse ionic layers around a colloidal particle. A high absolute value of zeta potential ($|\zeta| > 30$ mV) indicates electrostatic stability. When $\zeta \rightarrow 0$, coagulation occurs.

Stability Criterion:

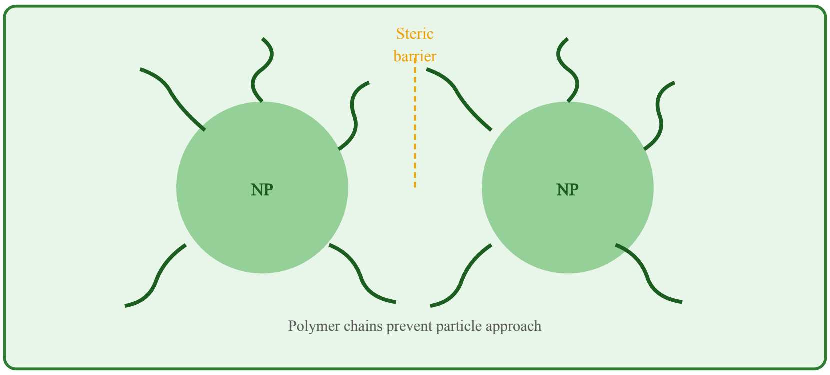
```
|ζ| > 30 mV → stable colloid |ζ| < 15 mV → unstable / coagulates
```

6.3 Protective Colloids

Lyophilic colloids (gelatin, starch, gum arabic) are added to lyophobic sols to stabilize them. They coat the particles and prevent coagulation. The amount of gold sol protected by 1 g of the protective colloid is the **Gold Number**. Lower gold number = better protective colloid.

Protective Colloid	Gold Number
Gelatin	0.005–0.01
Gum arabic	0.1–0.2
Starch	25
Haemoglobin	0.03

FIG 6.1 – STERIC STABILIZATION BY SURFACTANT / POLYMER



Polymer chains adsorbed on particle surface provide steric repulsion against aggregation

6.4 Coagulation and Flocculation

Coagulation is the process of aggregation of colloidal particles, caused by:

- Addition of electrolytes (neutralise surface charge)
- Heating or boiling
- Mixing of oppositely charged sols
- Persistent dialysis

Hardy–Schulze Rule: The coagulating power of an ion increases with its valence. For a negative sol: $Al^{3+} > Ca^{2+} > Na^{+}$. For a positive sol: $PO_4^{3-} > SO_4^{2-} > Cl^{-}$.

7 Introduction to Nanomaterials

7.1 Definition

Nanomaterials are materials with at least one dimension in the range of **1–100 nanometres (nm)**. At the nanoscale, materials exhibit unique physical, chemical, optical, electrical, and magnetic properties that differ markedly from their bulk counterparts.

> **Scale reference:**
>
> `1 nm = 10⁻⁹ m Human hair: ~80,000 nm wide Nanoparticle range: 1-100 nm Atom (H): ~0.1 nm`

7.2 Classification of Nanomaterials

Type	Description	Example
0D (Quantum dots)	All dimensions at nanoscale	Fullerene C_{60}, Au nanoparticles
1D (Nanowires/rods)	One dimension > nanoscale	Carbon nanotubes, TiO_2 nanorods
2D (Nanosheets)	Two dimensions > nanoscale	Graphene, MoS_2 sheets
3D (Bulk nanostructured)	Composed of nanocrystals	Nanocomposites

7.3 Nanomaterials vs. Bulk Materials

Property	Bulk Material	Nanomaterial
Surface area	Low	Very high ($SA/V \propto 1/r$)
Melting point	Fixed (high)	Significantly lower
Reactivity	Low to moderate	Highly reactive
Optical properties	Fixed colour	Size-dependent colour (SPR)
Electrical properties	Bulk conductivity	Quantum confinement effects
Strength	Normal	Often significantly higher

7.4 Unique Properties of Nanomaterials

A. Surface Plasmon Resonance (SPR)

Metal nanoparticles (Au, Ag) exhibit brilliant colours due to collective oscillation of electrons when irradiated with light. Gold nanoparticles appear red (not yellow like bulk gold) at ~20 nm.

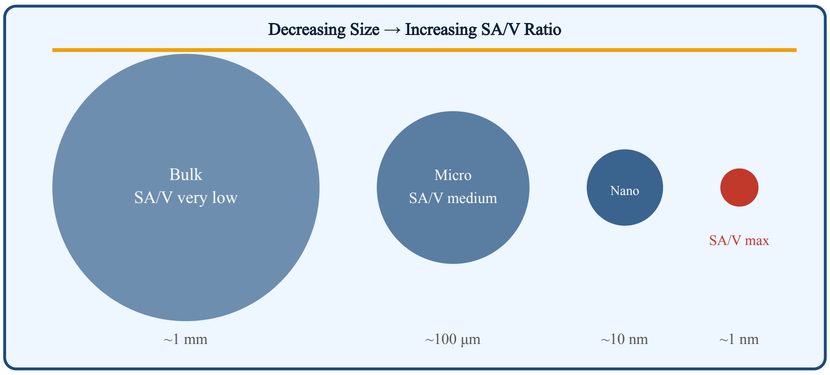
B. Quantum Confinement

When particle size approaches the de Broglie wavelength of electrons, quantum mechanical effects dominate. Energy levels become discrete, leading to tunable optical and electronic properties in semiconductor quantum dots.

C. High Surface-to-Volume Ratio

A 1 nm particle has ~50% of its atoms on the surface compared to ~0.1% for a 1 μm particle. This drastically increases reactivity and catalytic efficiency.

FIG 7.1 – SURFACE-TO-VOLUME RATIO VS. PARTICLE SIZE



As particle size decreases to nanoscale, surface-to-volume ratio increases dramatically

8 Preparation of Nanometals

8.1 Chemical Methods

A. Chemical Reduction Method

Metal salt solutions are reduced by reducing agents to form metal nanoparticles. Stabilizing agents (capping agents) are added simultaneously to prevent aggregation.

Gold Nanoparticles (Turkevich Method):

$HAuCl_4$ + Sodium citrate → Au nanoparticles (15–20 nm, red) Citrate acts as both reducing and capping agent

Silver Nanoparticles:

$AgNO_3 + NaBH_4 \rightarrow$ Ag nanoparticles $+ NaNO_3 + BH_3$ ($NaBH_4$ = sodium borohydride, reducing agent)

B. Sol-Gel Method

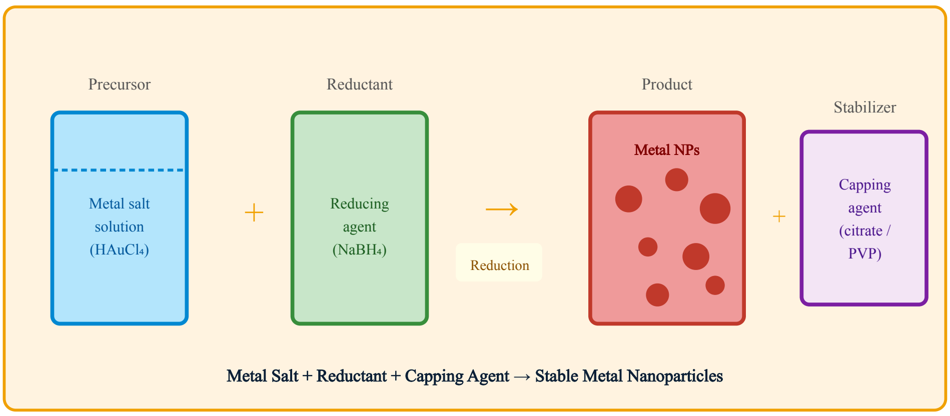
Involves hydrolysis and condensation of metal alkoxide precursors to form a gel, which is dried and calcined to get nanoparticles.

1. Metal alkoxide (e.g., TEOS for SiO_2) dissolved in alcohol.
2. Water added → hydrolysis → Si–OH groups form.
3. Condensation → Si–O–Si network (gel).
4. Drying (xerogel) or supercritical drying (aerogel).
5. Calcination at high temperature → nanoparticles.

C. Coprecipitation Method

Used for metal oxide nanoparticles. Metal salts are dissolved and precipitated simultaneously by adding a base. Widely used for magnetic nanoparticles (Fe_3O_4).

FIG 8.1 – CHEMICAL SYNTHESIS OF METAL NANOPARTICLES



Chemical reduction route for metal nanoparticle synthesis

8.2 Biological (Green) Synthesis of Nanometals

Biological synthesis uses organisms (plants, bacteria, fungi, algae) as bioreactors for nanoparticle synthesis. It is eco-friendly, non-toxic, and cost-effective.

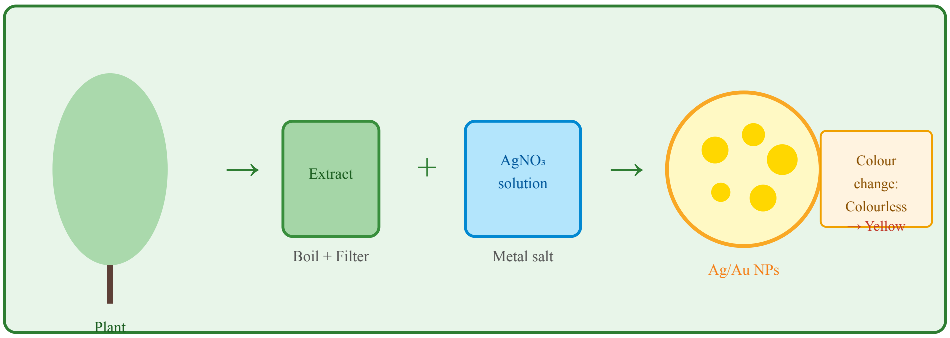
Plant-Mediated Synthesis

1. Plant leaf extract is prepared (boiling, filtering).
2. Extract mixed with metal salt solution (e.g., $AgNO_3$ or $HAuCl_4$).
3. Phytochemicals (flavonoids, terpenoids, alkaloids) act as reducing and capping agents.
4. Colour change confirms nanoparticle formation ($AgNO_3$: colourless → yellow/brown).
5. Nanoparticles are centrifuged and characterised by TEM, XRD, UV-Vis.

Example – Green Synthesis of Silver Nanoparticles:

```
AgNO₃ + Neem leaf extract → Ag nanoparticles (10–50 nm) Reducing agent: flavonoids / terpenoids present in extract Capping: same phytochemicals stabilize the surface
```

FIG 8.2 – BIOLOGICAL SYNTHESIS OF NANOPARTICLES



Green/biological synthesis: plant extract reduces metal salt → stable nanoparticles

9 Preparation of Nanometal Oxides

9.1 Important Nanometal Oxides

Nanometal Oxide	Application
TiO_2 (Titanium dioxide)	Photocatalysis, sunscreen, solar cells
ZnO (Zinc oxide)	Antimicrobial, UV protection, sensors
Fe_3O_4 (Iron oxide)	MRI contrast agents, drug delivery
CeO_2 (Cerium oxide)	Catalysis, antioxidant therapy
Al_2O_3 (Alumina)	Abrasives, thermal barrier coatings
CuO (Copper oxide)	Antibacterial, solar energy

9.2 Chemical Preparation of TiO_2 Nanoparticles (Sol-Gel)

1. Titanium isopropoxide (TTIP) dissolved in isopropanol under nitrogen atmosphere.
2. Controlled addition of water + acid catalyst (HNO_3/HCl).
3. Hydrolysis: $Ti(OPr)_4 + 4H_2O \longrightarrow Ti(OH)_4 + 4PrOH$
4. Condensation: $nTi(OH)_4 \longrightarrow (TiO_2)_n + nH_2O$ (gel forms)
5. Gel dried at 100–120°C ⟶ amorphous TiO_2
6. Calcination at 400–500°C ⟶ crystalline anatase TiO_2 nanoparticles.

Sol-Gel Reactions for TiO_2:

Hydrolysis: $Ti(OR)_4 + 4H_2O \rightarrow Ti(OH)_4 + 4ROH$ Condensation: $Ti(OH)_4 \rightarrow TiO_2 + 2H_2O$ (R = isopropyl; OR = isopropoxide group)

9.3 Preparation of Fe_3O_4 Nanoparticles (Coprecipitation)

1. Dissolve $FeCl_2$ and $FeCl_3$ in water (molar ratio Fe^{2+}:Fe^{3+} = 1:2).
2. Add NH_4OH solution dropwise under vigorous stirring at 80°C.
3. Black precipitate of Fe_3O_4 forms immediately.
4. Wash with water, centrifuge, dry.
5. Size can be controlled by pH, temperature, and stirring speed.

Coprecipitation Reaction:

$Fe^{2+} + 2Fe^{3+} + 8OH^- \rightarrow Fe_3O_4 + 4H_2O$

9.4 Hydrothermal Method

Metal salt precursors dissolved in water are placed in a sealed autoclave at temperatures 130–250°C and high pressure. The confined high-temperature conditions allow crystallisation of metal oxide nanoparticles with controlled morphology. Used for ZnO nanorods, TiO_2 nanotubes.

9.5 Biological Preparation of ZnO Nanoparticles

1. Prepare aqueous extract from Aloe vera / Azadirachta indica (neem) leaves.
2. Add zinc acetate [$Zn(CH_3COO)_2$] solution to the extract.
3. Heat mixture at 60°C with stirring for 2 hours.
4. Phytochemicals reduce Zn^{2+} and cap the nanoparticles.
5. Centrifuge, wash, calcine at 400°C → crystalline ZnO NPs.

Characterization Techniques: X-Ray Diffraction (XRD) – crystal structure and particle size via Scherrer equation; TEM – morphology; UV-Vis spectroscopy – optical properties; FTIR – surface functional groups; DLS – particle size distribution.

10 Stabilization of Nanomaterials

10.1 Need for Stabilization

Nanoparticles have high surface energy and tend to agglomerate to minimise surface area. Agglomeration destroys the unique properties of nanomaterials. Stabilization prevents aggregation and ensures long-term stability of nanoparticle dispersions.

10.2 Stabilization Mechanisms

A. Electrostatic Stabilization

Surface charge is introduced on nanoparticles (by adsorption of ions or surface functionalisation). Like-charged particles repel each other, preventing aggregation. Zeta potential $|\zeta| > 30$ mV ensures stability.

B. Steric Stabilization

Polymers (PVP, PEG, PVA) or surfactants are adsorbed onto nanoparticle surfaces. The polymer chains provide a physical barrier, preventing particle approach. This is particularly effective in organic solvents where electrostatic stabilization fails.

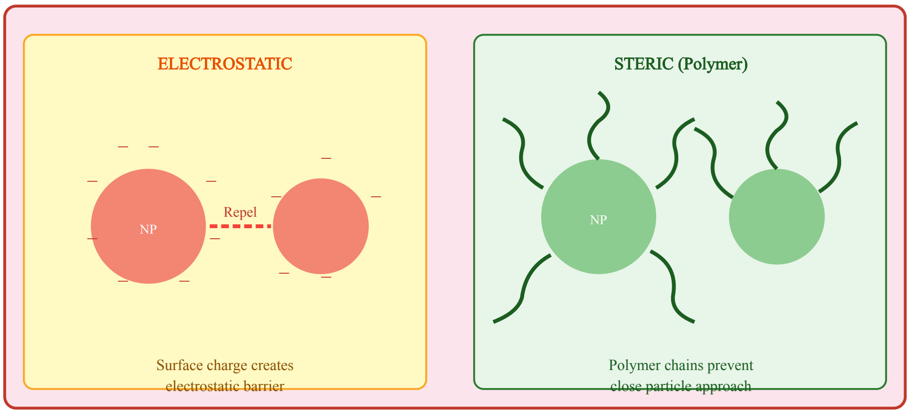
C. Electrosteric Stabilization

Combination of both electrostatic and steric mechanisms using charged polymers (polyelectrolytes). Provides robust stability over a wide pH and ionic strength range.

10.3 Common Stabilizing Agents

Stabilizing Agent	Type	Used For
Sodium citrate	Electrostatic	Gold, silver NPs
PVP (Polyvinylpyrrolidone)	Steric	Ag, Cu, Pt NPs
PEG (Polyethylene glycol)	Steric	Biomedical NPs
CTAB (surfactant)	Electrostatic + Steric	Au nanorods
Oleic acid	Steric	Fe_3O_4, CdSe NPs
Thiol (-SH) groups	Covalent capping	Au NPs
SDS (sodium dodecyl sulfate)	Ionic surfactant	Metal NPs

FIG 10.1 – ELECTROSTATIC VS STERIC STABILIZATION OF NANOPARTICLES



Two principal mechanisms for nanoparticle stabilization

11 Adsorption Isotherms

11.1 Definition of Adsorption Isotherm

An **adsorption isotherm** is a graph that shows the relationship between the amount of adsorbate adsorbed per unit mass of adsorbent (x/m) and the equilibrium pressure (or concentration) of the adsorbate at **constant temperature**.

11.2 Freundlich Adsorption Isotherm

Proposed by Herbert Freundlich (1909), it is an empirical equation describing adsorption of gases on solid surfaces.

Freundlich Equation:

```
x/m = k · P^(1/n) where: x/m = amount adsorbed per gram of adsorbent P = equilibrium pressure k, n = Freundlich constants (n > 1)
```

Linearised form (log-log plot):

```
log(x/m) = log k + (1/n) · log P Slope = 1/n, Intercept = log k
```

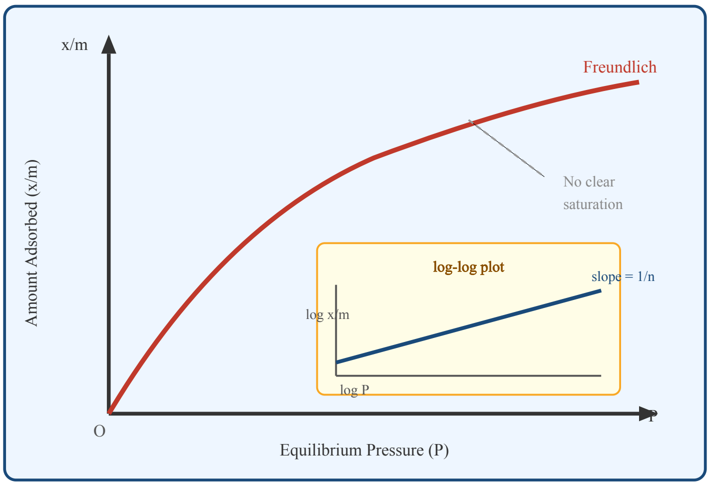
Verification of Freundlich Isotherm

Plot log(x/m) vs log(P). A straight line confirms the isotherm. The slope gives 1/n and the y-intercept gives log k.

Limitations

- Empirical – lacks theoretical basis.
- Does not apply at very high pressures (x/m does not reach saturation).
- Only valid over a limited pressure range.

FIG 11.1 – FREUNDLICH ADSORPTION ISOTHERM GRAPH



Freundlich isotherm (left) and its linearised log-log form (inset)

11.3 Langmuir Adsorption Isotherm

Proposed by Irving Langmuir (1916) based on theoretical considerations. Assumes: (1) monolayer adsorption, (2) uniform surface (homogeneous), (3) no lateral interactions between adsorbed molecules.

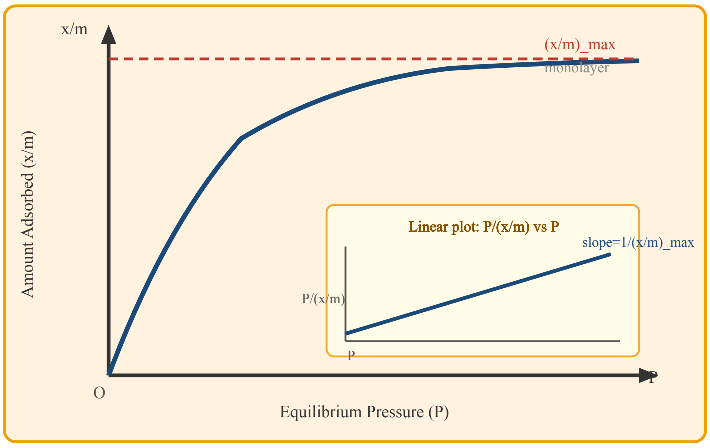
Langmuir Equation:

`θ = (K · P) / (1 + K · P) or equivalently: x/m = (x/m)_max · (KP)/(1 + KP) where: θ = fraction of surface covered K = Langmuir constant (related to adsorption energy) P = equilibrium pressure (x/m)_max = maximum adsorption (monolayer capacity)`

Linearised form:

`P/(x/m) = P/(x/m)_max + 1/(K·(x/m)_max) Plot P/(x/m) vs P → slope = 1/(x/m)_max, intercept = 1/(K·(x/m)_max)`

FIG 11.2 – LANGMUIR ADSORPTION ISOTHERM



Langmuir isotherm (approaches monolayer saturation) and linear plot

11.4 Freundlich vs Langmuir Comparison

Feature	Freundlich	Langmuir
Basis	Empirical	Theoretical
Surface type	Heterogeneous	Homogeneous
Layers	Multilayer	Monolayer
Saturation	Not predicted	Predicted (asymptote)
Equation	$x/m = kP^{(1/n)}$	$\theta = KP/(1+KP)$
Linear plot	log(x/m) vs log P	P/(x/m) vs P

12 BET Equation and Surface Area Measurement

12.1 Introduction to BET Theory

The **BET equation** was developed by **Brunauer, Emmett, and Teller** (1938). It is an extension of Langmuir theory to **multilayer adsorption**. BET theory assumes that the Langmuir equation applies to each adsorbed layer independently.

12.2 BET Equation

BET Equation (No derivation required):

$1 / [V((P_0/P) - 1)] = (C - 1)(P/P_0) / (V_m \cdot C) + 1/(V_m \cdot C)$ where: P = equilibrium pressure of adsorbate P_0 = saturation vapour pressure of adsorbate V = volume of gas adsorbed at pressure P V_m = volume of gas to form monolayer (monolayer capacity) C = BET constant (related to enthalpy of adsorption)

Linear Plot:

Plot $1/[V(P_0/P - 1)]$ vs P/P_0 Slope = $(C-1)/(V_m \cdot C)$ Intercept = $1/(V_m \cdot C)$ → $V_m = 1/(\text{Slope} + \text{Intercept})$ → $C = (\text{Slope}/\text{Intercept}) + 1$

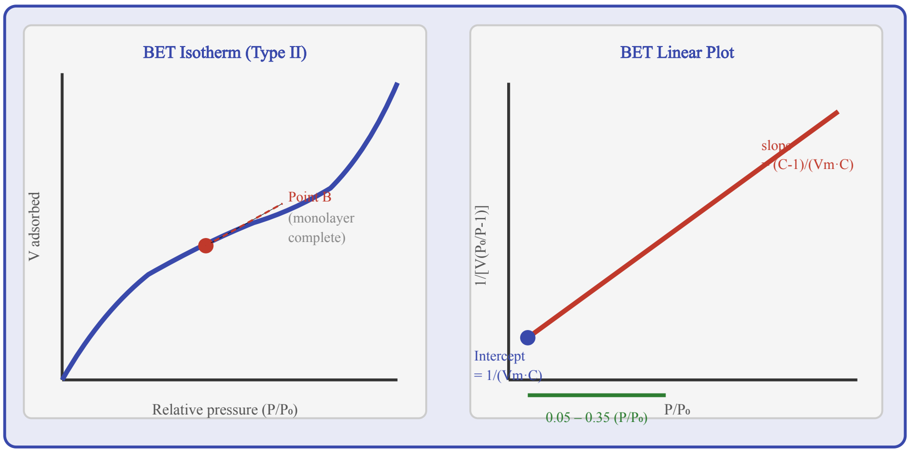
BET Surface Area:

$S_{BET} = (V_m \cdot N_A \cdot a_m) / (22400)$ where a_m = area of one adsorbate molecule (N_2: 0.162 nm^2) N_A = Avogadro's number = 6.022×10^{23} mol^{-1}

12.3 Physical Significance

- BET surface area is the gold standard for measuring specific surface area of porous materials.
- Nitrogen gas at 77 K (liquid N_2 temperature) is the most common adsorbate.
- Essential for characterising catalysts, adsorbents, activated carbon, zeolites.
- P/P_0 range for BET analysis: 0.05–0.35.

FIG 12.1 – BET ISOTHERM (TYPE II) AND BET LINEAR PLOT



BET Type-II isotherm shows multilayer adsorption; linear BET plot used to determine Vm and C

Applications of BET Analysis: Catalyst characterization, quality control of activated carbon, zeolite surface area, pharmaceutical powder characterization, soil science, fuel cell membrane studies.

13 Applications of Colloids and Nanomaterials

13.1 Applications of Colloids

Field	Application	Colloid Type
Medicine	Blood is a colloidal sol; collargol (Ag sol) antiseptic	Sol
Food	Milk (fat in water), butter, ice cream, mayonnaise	Emulsion/Foam
Industry	Rubber (latex), paint, ink, adhesives	Sol/Emulsion
Pollution control	Cottrell precipitator removes colloidal smoke/dust	Aerosol
Photography	AgBr particles in gelatin on film	Solid sol
Water treatment	Alum coagulates suspended particles	Sol
Detergency	Soap micelles remove grease	Micelle

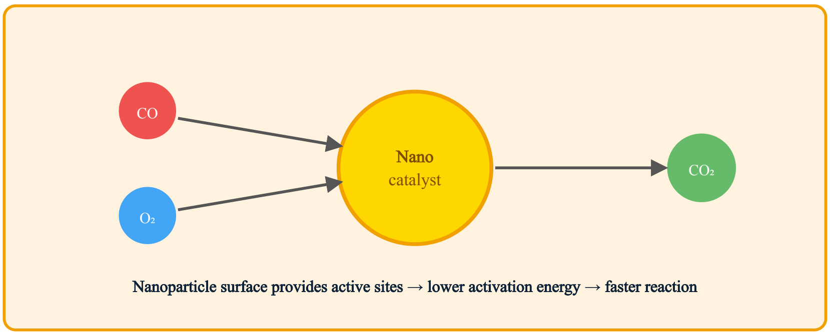
13.2 Applications of Nanomaterials

A. Catalysis

Nanoparticles provide enormous surface area for catalytic reactions. They lower activation energy and increase reaction rate and selectivity.

- Pt/Pd nanoparticles in automotive catalytic converters (oxidize CO to CO_2).
- TiO_2 nanoparticles: photocatalytic degradation of pollutants.
- Au nanoparticles: CO oxidation at room temperature.
- Fe nanoparticles in Fischer-Tropsch synthesis.

FIG 13.1 – CATALYSIS BY NANOPARTICLES



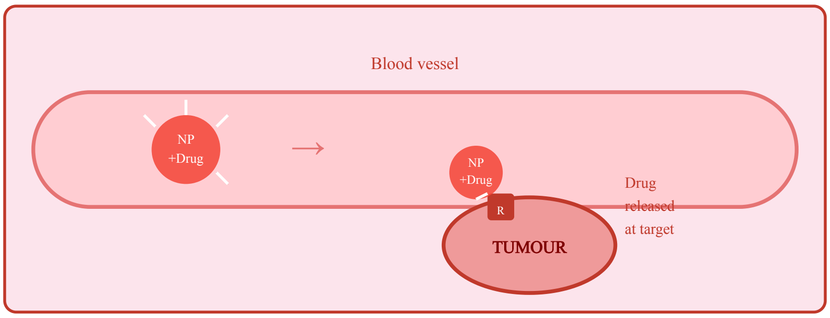
Nanoparticle catalyst facilitates $CO + \frac{1}{2}O_2 \rightarrow CO_2$ at lower activation energy

B. Medicine and Drug Delivery

- **Targeted drug delivery:** Polymer nanoparticles carry anticancer drugs to tumour sites (EPR effect).

- **MRI contrast agents:** Fe_3O_4 nanoparticles enhance MRI imaging.
- **Antibacterial:** Silver nanoparticles destroy bacterial membranes — used in wound dressings, coatings.
- **Gene therapy:** Gold nanoparticles as gene delivery vectors.
- **Diagnostic biosensors:** Au NP-based lateral flow assays (e.g., rapid antigen tests).

FIG 13.2 – TARGETED DRUG DELIVERY BY NANOPARTICLES



Nanoparticles with surface ligands recognise tumour receptors and release drug at site

C. Sensors

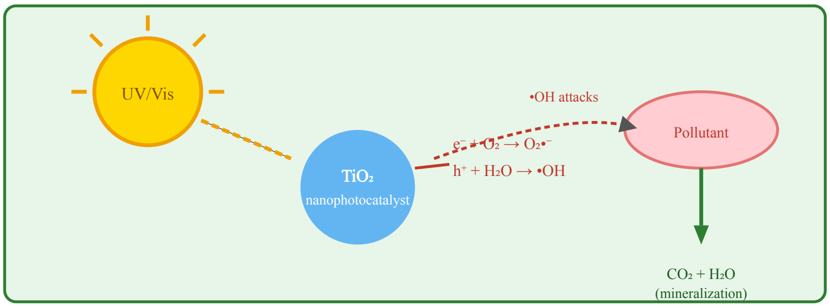
Nanoparticles enhance sensor sensitivity due to high surface area and unique optical/electrical properties.

- **Glucose biosensor:** Gold nanoparticles on electrode detect glucose via enzymatic reaction.
- **Gas sensor:** SnO_2 nanoparticles detect reducing gases (CO, H_2) through resistance change.
- **Colorimetric sensor:** AgNPs change colour upon exposure to specific analytes.
- **SERS (Surface Enhanced Raman Scattering):** Au/Ag NPs amplify Raman signal up to 10^8 times.

D. Environmental Applications

- **Water purification:** TiO_2 nanoparticles photocatalytically degrade organic pollutants; Fe nanoparticles remove heavy metals.
- **Air purification:** Photocatalytic coatings on buildings decompose NO_x, SO_2, VOCs.
- **Groundwater remediation:** Zero-valent iron nanoparticles (nZVI) degrade chlorinated solvents.
- **Antibacterial water treatment:** Ag and Cu nanoparticles in water filters.

FIG 13.3 – ENVIRONMENTAL APPLICATION: PHOTOCATALYTIC DEGRADATION



14 Questions & Answers

Part A – 2-Mark Questions

2 Marks

Q1. Define adsorption and distinguish it from absorption.

Adsorption is the process of accumulation of molecules of a substance (adsorbate) on the surface of another substance (adsorbent). It is a surface phenomenon where adsorbate molecules concentrate only at the surface. **Absorption**, in contrast, is the process in which a substance uniformly penetrates into the bulk of another substance (e.g., a sponge absorbing water). The key distinction is that adsorption is superficial while absorption is volumetric. An important industrial example: activated charcoal adsorbs gases on its surface; cotton absorbs water into its bulk. The term *sorption* encompasses both phenomena.

2 Marks

Q2. What is the Tyndall effect? Give one example.

The **Tyndall effect** is the scattering of a beam of light by the particles in a colloidal solution, making the path of the light beam visible. When a ray of light is passed through a colloidal sol, the colloidal particles scatter light in all directions, so the beam becomes visible as a bright cone (Tyndall cone). This occurs because the particle size in a colloid (1–1000 nm) is comparable to the wavelength of visible light. True solutions do not show this effect because the solute particles are too small. **Example:** The path of a torch beam becomes visible when passed through a fog or smoke — both are colloidal systems. This effect is absent in a true solution of NaCl.

2 Marks

Q3. State the Hardy-Schulze rule with an example.

The **Hardy-Schulze rule** states that the coagulating power of an electrolyte ion is directly proportional to the valency (charge) of the ion responsible for coagulation. In other words, ions with higher valence are more effective at precipitating (coagulating) a colloidal sol. For a *negatively charged sol* (e.g., As_2S_3 sol), the coagulating power of cations follows: $Al^{3+} > Ca^{2+} > Na^+$. For a *positively charged sol* (e.g., $Fe(OH)_3$ sol), the coagulating power of anions follows: $PO_4^{3-} > SO_4^{2-} > Cl^-$. **Example:** Alum $[Al_2(SO_4)_3]$ is highly effective in water treatment because Al^{3+} has high coagulating power for negatively charged clay/mud particles.

2 Marks

Q4. What are nanomaterials? State their size range.

Nanomaterials are materials in which at least one dimension is in the nanometre scale, typically between **1 nm and 100 nm** (1 nm = 10^{-9} m). At this size scale, materials exhibit unique properties

— optical, magnetic, electrical, and chemical — that are distinctly different from their bulk counterparts. These novel properties arise because of (i) the very high surface-to-volume ratio (more atoms exposed at the surface), and (ii) quantum mechanical effects (quantum confinement). Examples include gold nanoparticles (appear red/purple, not gold), carbon nanotubes (stronger than steel), and ZnO nanoparticles (antibacterial). The field is called *nanotechnology* and covers 0D (quantum dots), 1D (nanowires), 2D (nanosheets), and 3D nanostructured materials.

Part B – 5-Mark Questions

5 Marks

Q5. Explain the Freundlich adsorption isotherm with the equation and graph.

Freundlich Adsorption Isotherm

The Freundlich adsorption isotherm is an empirical equation proposed by Herbert Freundlich (1909) that describes the relationship between the amount of adsorbate adsorbed per unit mass of adsorbent and the equilibrium pressure at constant temperature.

Equation: $x/m = k \cdot P^{(1/n)}$
where x = mass of adsorbate, m = mass of adsorbent, P = equilibrium pressure, k and n are Freundlich constants ($n > 1$).

Linearised form: Taking logarithm: $\log(x/m) = \log k + (1/n) \log P$. Plotting $\log(x/m)$ vs $\log P$ gives a straight line with slope $1/n$ and intercept $\log k$.

Features of the graph: The plot of x/m vs P is a curve that rises continuously with pressure and does not show a saturation plateau. This is a limitation — it does not predict a maximum adsorption capacity.

Significance: The constant $1/n$ (between 0 and 1) indicates the intensity of adsorption. When $1/n = 1$, the equation becomes linear (Henry's law). For $1/n < 1$, the surface has high-affinity sites; as sites fill, the affinity decreases.

Limitations: (i) It is purely empirical with no theoretical basis. (ii) It is valid only over a limited pressure range. (iii) It does not predict adsorption saturation. (iv) It cannot account for multilayer adsorption quantitatively.

Applications: Used to study gas-solid adsorption, purification of water using activated charcoal, and soil science.

5 Marks

Q6. Explain the Langmuir adsorption isotherm with theory and graph.

Langmuir Adsorption Isotherm

Proposed by Irving Langmuir (1916), this isotherm is based on the following theoretical assumptions: (1) adsorption occurs only as a monolayer; (2) the surface is homogeneous (all sites have equal energy); (3) there are no lateral interactions between adsorbed molecules; (4) at equilibrium, the rate of adsorption equals the rate of desorption.

Derivation idea: At equilibrium: rate of adsorption = $k_1P(1-\theta)$ = rate of desorption = $k_2\theta$, where θ = fraction of surface covered. Solving: $\theta = KP/(1+KP)$, $K = k_1/k_2$.

Equation: $x/m = (x/m)_{max} \cdot KP/(1+KP)$
At low P: $x/m \propto P$ (linear region, Henry's law). At high P: $x/m \rightarrow (x/m)_{max}$ (saturation — monolayer complete).

Linearised form: $P/(x/m) = P/(x/m)_{max} + 1/[K(x/m)_{max}]$. Plot $P/(x/m)$ vs P → straight line; slope = $1/(x/m)_{max}$; intercept = $1/[K(x/m)_{max}]$.

Graph: The x/m vs P curve rises steeply at low P, then flattens and asymptotically approaches $(x/m)_{max}$ — indicating monolayer completion.

Assumptions & Limitations: Assumes uniform surface (rarely true in practice). Fails at high pressure if multilayer adsorption occurs. The concept of fixed active sites may not hold for all adsorbents.

5 Marks

Q7. Explain micelle formation with diagram and applications.

Micelle Formation

A micelle is an organized aggregate of surfactant (surface-active) molecules that spontaneously forms in a solution when the surfactant concentration exceeds the **Critical Micelle Concentration (CMC)**.

Structure of Surfactant: Each surfactant molecule consists of a hydrophilic (water-loving) polar head group and a hydrophobic (water-hating) nonpolar hydrocarbon tail.

Mechanism:
(i) Below CMC: Surfactant monomers spread at the air-water interface as a monolayer, with tails pointing out of water and heads in water.
(ii) At CMC: The surface becomes saturated. Further increase in concentration leads to micelle formation in the bulk.
(iii) Above CMC: Spherical micelles form. The hydrophobic tails cluster inward (away from water) forming the core; the hydrophilic heads face outward (toward water), forming the shell.

Properties: Micelle diameter ~5–20 nm; interior resembles liquid hydrocarbon; CMC of sodium stearate ~10^{-3} mol/L. Above CMC, surface tension becomes constant.

Applications:
1. Detergency — grease trapped in hydrophobic core, removed by water.
2. Drug delivery — hydrophobic drugs solubilized in micelle interior.
3. Emulsification — stabilizes oil-water emulsions.
4. Cosmetics — shampoos, conditioners, cleansers.
5. Nanoparticle synthesis — reverse micelles as nanoreactors.
6. Extraction of essential oils and aromatic compounds.

5 Marks

Q8. Explain the preparation of nanomaterials by chemical and biological methods.

Preparation of Nanomaterials

A. Chemical Methods:
(i) Chemical Reduction: Metal salt solutions are reduced using reducing agents ($NaBH_4$, sodium citrate) in the presence of a stabilizing agent (capping agent) to prevent aggregation. Example: $HAuCl_4$ + sodium citrate → Au nanoparticles (15–20 nm, red colour); $AgNO_3 + NaBH_4$ → Ag

nanoparticles.

(ii) Sol-Gel Method: Metal alkoxide is hydrolysed and condensed to form a gel, dried, and calcined. Used for TiO_2, SiO_2, ZnO nanoparticles. Provides precise control over size and morphology.

(iii) Coprecipitation: Metal salts precipitated simultaneously by base. Example: $FeCl_2 + FeCl_3 + NH_4OH \rightarrow Fe_3O_4$ nanoparticles (magnetic).

B. Biological (Green) Methods:

Plants, bacteria, fungi, and algae are used as bioreactors. For plant-mediated synthesis: leaf extract is prepared; mixed with metal salt solution; phytochemicals (flavonoids, terpenoids) act as reducing and capping agents; nanoparticles form as evidenced by colour change. Example: Neem extract + $AgNO_3 \rightarrow$ Ag nanoparticles (10–50 nm).

Advantages of biological method: Eco-friendly (no toxic reagents), cost-effective, biocompatible products, single-step synthesis, scalable for industrial production. Nanoparticles are already surface-capped by natural phytochemicals, making them suitable for biomedical applications directly.

5 Marks

Q9. Write about the applications of nanomaterials in medicine, sensors, and environment.

Applications of Nanomaterials

A. Medicine:

(i) *Drug Delivery:* Polymer nanoparticles (PLGA, chitosan) encapsulate drugs and deliver them specifically to tumour sites via EPR (enhanced permeability and retention) effect, reducing side effects.

(ii) *Diagnostic Imaging:* Fe_3O_4 nanoparticles used as MRI contrast agents; gold nanoparticles used in X-ray CT imaging.

(iii) *Antibacterial:* Silver nanoparticles disrupt bacterial cell membranes → used in wound dressings, surgical instruments, hospital coatings.

(iv) *Cancer therapy:* Gold nanoshells/nanorods used in photothermal therapy — absorb near-IR light and convert to heat, killing cancer cells selectively.

B. Sensors:

(i) Gold nanoparticles modified electrodes → highly sensitive electrochemical biosensors for glucose, DNA, proteins.

(ii) SnO_2 nanoparticle gas sensors → detect CO, H_2, ethanol at ppm levels through resistance changes.

(iii) Colourimetric sensors based on Ag/Au NPs → naked-eye detection of heavy metals (Pb^{2+}, Hg^{2+}) in water.

(iv) SERS-based sensors → ultra-sensitive pathogen detection (COVID-19 antigens).

C. Environmental Applications:

(i) TiO_2 nanoparticles photocatalytically degrade organic dyes, pesticides, and pharmaceuticals in water under UV/visible light.

(ii) Zero-valent iron nanoparticles (nZVI) reduce chlorinated solvents and heavy metals in contaminated groundwater.

(iii) Ag and Cu nanoparticle-loaded filters purify drinking water by killing pathogens.

(iv) Photocatalytic coatings on building materials decompose NO_x and SO_2 from polluted air.

Numerical Problems

Problem 1 – Freundlich Isotherm Calculation

Q: The following data were obtained for adsorption of gas on a solid at 300 K. Verify Freundlich isotherm and determine constants k and n.

P (atm)	1	2	4	8	16

x/m (g/g)	0.20	0.28	0.40	0.56	0.80

Solution:

Take log on both sides: log(x/m) = log k + (1/n)log P

P	x/m	log P	log(x/m)
1	0.20	0.000	−0.699
2	0.28	0.301	−0.553
4	0.40	0.602	−0.398
8	0.56	0.903	−0.252
16	0.80	1.204	−0.097

Slope = Δlog(x/m) / Δlog P = (−0.097 − (−0.699)) / (1.204 − 0.000) = 0.602 / 1.204 ≈ **0.50 = 1/n → n = 2**

Intercept = −0.699 → log k = −0.699 → **k = 0.20**

∴ Freundlich equation: **x/m = 0.20 × P^(0.5)**

Problem 2 – Langmuir Isotherm Calculation

Q: The Langmuir constants for adsorption of CO on Ni are: (x/m)_max = 1.2 g/g, K = 0.5 atm^{-1}. Calculate x/m at P = 2 atm.

Solution:

Using Langmuir equation: x/m = (x/m)_max · KP / (1 + KP)

x/m = 1.2 × (0.5 × 2) / (1 + 0.5 × 2)

x/m = 1.2 × 1.0 / (1 + 1.0)

x/m = 1.2 / 2.0 = **0.60 g/g**

Quick Revision Sheet

Use this sheet in the last hour before your exam for rapid recollection.

Key Formulas

- Freundlich: $x/m = kP^{(1/n)}$
- $\log(x/m) = \log k + (1/n)\log P$
- Langmuir: $\theta = KP/(1+KP)$
- Linear Langmuir: $P/(x/m) = P/(x/m)_{max} + 1/[K(x/m)_{max}]$
- BET: $1/[V(P_0/P-1)] = (C-1)(P/P_0)/(V_mC) + 1/(V_mC)$
- $SA = V_m \times N_A \times a_m / 22400$

Adsorption Types

- Physisorption: van der Waals, 5–40 kJ/mol, multilayer, reversible, low T
- Chemisorption: chemical bond, 40–400 kJ/mol, monolayer, irreversible, high T
- Factors: surface area, T, P, nature of adsorbate/adsorbent

Colloids

- Size: 1–1000 nm
- Tyndall effect → scattered light visible
- Brownian motion → zig-zag random motion
- Electrophoresis → migration under electric field
- Hardy-Schulze: higher valency → higher coagulation power
- Gold number: lower = better protective colloid

Micelles

- CMC = Critical Micelle Concentration
- Hydrophilic head (polar) faces water
- Hydrophobic tail (nonpolar) forms inner core
- Application: detergency, drug delivery, emulsification
- Soap CMC ≈ 10^{-3} mol/L

Nanoparticle Preparation

- **Chemical reduction:** $HAuCl_4$ + citrate → Au NPs
- **Sol-gel:** $Ti(OR)_4 \rightarrow TiO_2$ NPs
- **Coprecipitation:** $FeCl_2 + FeCl_3 + NH_4OH \rightarrow Fe_3O_4$
- **Green synthesis:** Neem extract + $AgNO_3$ → Ag NPs
- Characterization: XRD, TEM, UV-Vis, FTIR, DLS

Stabilization

- Electrostatic: surface charge, $|\zeta| > 30$ mV stable
- Steric: PVP, PEG, oleic acid adsorbed on surface
- Electrosteric: combination (polyelectrolytes)
- Common agents: citrate (Au), PVP (Ag), PEG (biomedical)
- Thiol groups: covalent capping for Au NPs

Nanomaterial Applications

- **Catalysis:** Pt/Pd in catalytic converters, TiO_2 photocatalysis
- **Medicine:** targeted drug delivery, MRI (Fe_3O_4), Ag antibacterial
- **Sensors:** Au NP electrodes, SnO_2 gas sensors, SERS
- **Environment:** TiO_2 pollutant degradation, nZVI groundwater

Important Comparisons

- Freundlich: empirical, heterogeneous, multilayer
- Langmuir: theoretical, homogeneous, monolayer
- BET: multilayer, gives surface area, linear P/P_0 = 0.05–0.35
- Colloid vs True solution: size, Tyndall, stability
- Nano vs Bulk: SA/V, melting pt, colour, reactivity

MUST REMEMBER REACTIONS:

Peptization ($Fe(OH)_3$ sol): $Fe(OH)_3$ ppt + $FeCl_3$ → $Fe(OH)_3$ colloidal sol (Fe^{3+} adsorbed) Green Ag NPs: $AgNO_3$ + Plant extract (flavonoids) → Ag nanoparticles TiO_2 sol-gel: $Ti(OPr)_4 + H_2O$ → $Ti(OH)_4$ → TiO_2 (after calcination) Fe_3O_4 coprecipitation: $Fe^{2+} + 2Fe^{3+} + 8OH^- \rightarrow Fe_3O_4 + 4H_2O$ Photocatalysis: $TiO_2 + h\nu \rightarrow e^- + h^+$; $h^+ + H_2O \rightarrow \bullet OH$; $\bullet OH$ + pollutant → $CO_2 + H_2O$

⭐ **Exam Strategy:** For 2-mark Q ⟶ Definition + Example. For 5-mark Q ⟶ Definition + Theory + Equation + Graph sketch + Limitations + Applications. Always draw the diagram — it fetches bonus marks!

Textbook References

Textbooks

[T1] **Jain, P.C. and Jain, M.** – *Engineering Chemistry*, 16th Edition, Dhanpat Rai Publishing Company, New Delhi, 2013. *(Primary reference for Indian B.Tech curriculum – covers adsorption, colloids, surface chemistry fundamentally.)*

[T2] **Atkins, P., de Paula, J. and Keeler, J.** – *Atkins' Physical Chemistry*, 10th Edition, Oxford University Press, Oxford, 2010. *(Comprehensive treatment of adsorption isotherms, surface thermodynamics, and BET theory.)*

Reference Books

[R1] **Taylor, H.F.W.** – *Cement Chemistry*, 2nd Edition, Thomas Telford Publishing, 1997. *(Surface chemistry aspects of cement hydration; C-S-H gel formation.)*

[R2] **Shaw, D.J.** – *Introduction to Colloid and Surface Chemistry*, 4th Edition, Butterworth-Heinemann, Oxford, 1992. *(Classic text on colloidal systems, double layer theory, coagulation, and stabilization.)*

[R3] **Billmeyer, Fred W., Jr.** – *Textbook of Polymer Science*, 3rd Edition, Wiley-Interscience, New York, 1984. *(Steric stabilization by polymers; polymer adsorption at surfaces.)*

[R4] **Cao, G.** – *Nanostructures and Nanomaterials: Synthesis, Properties and Applications*, Imperial College Press, London, 2004.

[R5] **Ramsden, J.J.** – *Nanotechnology: An Introduction*, Elsevier, 2011.

Web Resources

- NPTEL (nptel.ac.in) – Engineering Chemistry Video Lectures by IIT Professors
- Virtual Labs (vlab.co.in) – Adsorption isotherm virtual experiments
- ChemLibreTexts (chem.libretexts.org) – Surface Chemistry modules
- American Chemical Society (acs.org) – Nanomaterials journal articles

Syllabus Mapping Summary:

Surface Chemistry Basics → Sections 1, 2 | Colloids → Sections 3, 4, 5, 6 | Nanomaterials → Sections 7, 8, 9, 10 | Adsorption Isotherms → Sections 11, 12 | Applications → Section 13 | Q&A → Section 14 | Revision → Section 15

Engineering Chemistry – Unit V

Surface Chemistry and Nanomaterials | B.Tech 1st Year